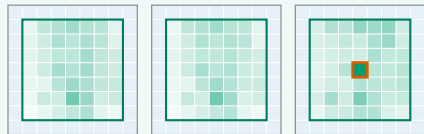


Inside the ShiftWM head: one patch i at horizon $k=10$ (trained model, held-out DROID)

1 Where does patch i come from?

$$\pi_{k,i}(j) = \text{softmax}_j (\mathbf{q}_{k,i}^\top \mathbf{k}_j / \sqrt{d} + \beta \mathbf{1}[j=(t,i)])$$

7×7 transport windows in the last 3 frames



31% of mass

27% of mass

42% of mass

weight π

— identity $(+\beta)$

2 Stay, or move

stay: $\mathbf{Z}_{t,i}$

$\times (1-g)$

move: $\mathbf{T}_{k,i} = \sum_j \pi(j) \mathbf{Z}_j$

$\times g$

gate $g_{k,i} = 0.99$

stay move

strips: 384-D features, top 12 PCs

3 Correct, predict

correction $\mathbf{r}_{k,i} = \mathbf{s} \odot \mathbf{W}_o \mathbf{h}_{k,i}$

$\times (1-g)$

forecast $\hat{\mathbf{Z}}_{k,i}$

$+$

true $\mathbf{Z}_{t+k,i}$

error: stay 1.65 $\rightarrow$ **ours 0.74**

all patches and steps in parallel